

StockKeep

Shop Inventory & Stock Tracking System

Final Submission Readiness Report

FINAL_SUBMISSION_READINESS_REPORT.pdf

Student Name:	Prince Ephraim Quarshie
Student ID:	22424601
Project Title:	StockKeep — AI Shop Inventory & Stock Tracking System
Institution:	University of Ghana
Department:	Computer Science
Module / Course:	CSCD602
Supervisor:	Prof. Solomon Mensah
Date:	13 August 2026

This report assesses the full submission package against what has actually been done, verified live, or still requires action. It supersedes an earlier draft of this report written before the code-level remediation pass described in *Technical_Debt_Plan.pdf* was carried out.

Contents

1	Summary	2
2	Complete	3
3	Requires Student Input	4
4	Requires Verification (by the student, before final upload)	5
5	Missing / Incomplete	6
6	Note on Repository State	7
7	Recommended Final Actions (in order)	8

1. Summary

Since the documentation package was first assembled, a full remediation pass was carried out: six technical debt items and defects were fixed in the source code, verified against a real production build, pushed to GitHub, and confirmed working on the live Vercel deployment. A seventh, more severe live-only defect (DEF-06 — the live database was completely non-functional) was discovered and mitigated during that same pass. This is a materially stronger submission than a documentation-only pass would have produced.

2. Complete

- **Project_Documentation.pdf** — master report, all 37 required sections, updated to reflect every fix below, consistent with all companion documents.
- **SRS.pdf** — standalone, NFR-02/AUTH-04/SEC-03/SEC-04/SEC-05 updated from “gap” to “resolved” with live evidence references.
- **Testing_Report.pdf** — three testing passes documented: Pass 1 (26 tests, dev server, found 5 defects), Pass 2 (11 tests, production build, confirmed all 5 fixed), Pass 3 (live deployment, found and fixed a 6th, more severe defect).
- **Technical_Debt_Plan.pdf** — 15-item register with live-verified Status column; 5 items RESOLVED, 1 MITIGATED (TD-08), 9 remain open with a phased roadmap.
- **User_Manual.pdf** — updated to remove stale caveats about the passcode field not working (it now does), and to describe the new rate-limiting and write-only secret fields.
- **Deployment_and_Source_Links.txt** — real live URL, real GitHub repository, real (initial) test passcode, all verified live immediately before finalising this package — not asserted from memory.
- **Supporting_Files/** — diagrams, requirements traceability matrix, and now three evidence logs: the original Pass 1 log, a production fix-verification log (Pass 2), and a live-deployment evidence log (Pass 3).
- **Six code fixes, actually shipped**, not just documented as a plan:
 1. Passcode hashing + `Settings.passcode` as the real login source of truth (was TD-01).
 2. `/api/seed` now requires authentication (was TD-02, a data-wipe exposure).
 3. Login rate limiting, 5 attempts/5 minutes (was TD-03).
 4. Secrets (Resend key, passcode) no longer returned by `GET /api/settings` (was TD-06/DEF-05).
 5. AI model switched to `gemini-flash-latest` after confirming the previous two model names were retired by Google (was TD-13/DEF-04) — verified live, genuine AI responses now returned.
 6. Root-caused and fixed the 500 error on invalid input: a Zod v4 API rename (`.errors` → `.issues`) that three catch blocks had missed (was DEF-01).
 7. **(Found and fixed live, not part of the original register)** The live database was entirely non-functional due to Vercel’s read-only serverless filesystem; mitigated via a `/tmp-copy` workaround, verified with a real write-then-read round trip on the live URL.
- All six/seven fixes were **pushed to GitHub and confirmed live**, not just applied locally.

3. Requires Student Input

- None of the previously-outstanding placeholder fields remain — name, ID, institution, department, module, and supervisor are filled in across all documents from the details you provided.
- The live passcode was updated from the earlier default (1234) to 557575 as part of this pass, and the login page's public passcode hint has been removed entirely. If you change it again, update `Deployment_and_Source_Links.txt` Section 3 so an examiner isn't locked out.

4. Requires Verification (by the student, before final upload)

- **UI/system-level testing.** All testing in this package is API-level (via `curl`) or live-endpoint testing; no rendered-browser testing was performed, because no browser automation tool was available in the environment used to prepare this package. Manually click through the live site's Dashboard, Add Item, Stock In/Out, Sales, Settings, and AI Chat before submission.
- **Screenshots.** Still genuinely absent — capture them from the now-working live site per `Supporting_Files/Screenshots/README_SCREENSHOTS_PENDING.txt` and insert into `User_Manual.pdf`.
- **Data persistence over time.** The `/tmp` workaround for the live database (Section on TD-08 in `Technical_Debt_Plan.pdf`) restores functionality but is not guaranteed to keep data indefinitely. If you plan to demo the app live, add your demo data shortly before presenting rather than relying on data added days earlier.
- **Usability, load/performance, and formal security testing** remain unexecuted, as documented in `Testing_Report.pdf`.

5. Missing / Incomplete

- **Automated test suite** (TD-07) — still zero automated tests. This is a genuine code-level gap, not a documentation omission; implementing it is a larger undertaking than this remediation pass and was correctly scoped out.
- **CSRF protection and security response headers** (TD-04, TD-05) — still open, lower priority than the six items fixed.
- **Durable database migration** (revised TD-08) — the live site works now, but a real managed database is still recommended before trusting it with real shop data long-term; this requires provisioning a database and setting Vercel environment variables, which needs the project owner's Vercel account access.
- **Frontend decomposition, pagination, and the other Version 2.0/Long-Term items** in `Technical_Debt_Plan.pdf` remain open by design — they were correctly deprioritised behind the higher-severity items actually fixed.

6. Note on Repository State

While this session’s changes were being made, a separate commit (24e6371, authored under the same GitHub account, message “updates”) appeared on `main` and bundled in this documentation package’s own output files (the `StudentID_StockKeep/` folder and its `.zip`) alongside a source-code change, plus an unrelated file reorganisation (`index.html` and related files moved into a `stitch screens/` folder). This was not an action taken deliberately as part of this remediation pass — it reflects the repository’s committed state at that point in time. **Recommended action:** before final submission, review whether you want the large documentation `.zip` and PDFs tracked inside the source repository itself (most courses expect the source repo to contain only source code, with documentation submitted separately to SAKAI) — if not, remove them from the repository in a follow-up commit; this does not need to block using the current submission package.

7. Recommended Final Actions (in order)

1. Capture real screenshots from the live site (now fully functional) and insert them into `User_Manual.pdf`.
2. Manually click through the core flows in a browser to confirm the UI itself works end-to-end (the API layer has been thoroughly verified; the rendered UI has not been separately tested).
3. Decide whether to keep `StudentID_StockKeep/` tracked inside the GitHub source repository or remove it (see the note above) — either is acceptable, but make it a deliberate choice.
4. If you want durable data for a live demo, consider migrating to a managed database (TD-08) beforehand — optional, but recommended if the demo matters for grading.
5. Re-zip the `StudentID_StockKeep/` folder as `StudentID_StockKeep.zip` (already done as part of this package) and upload to SAKAI.